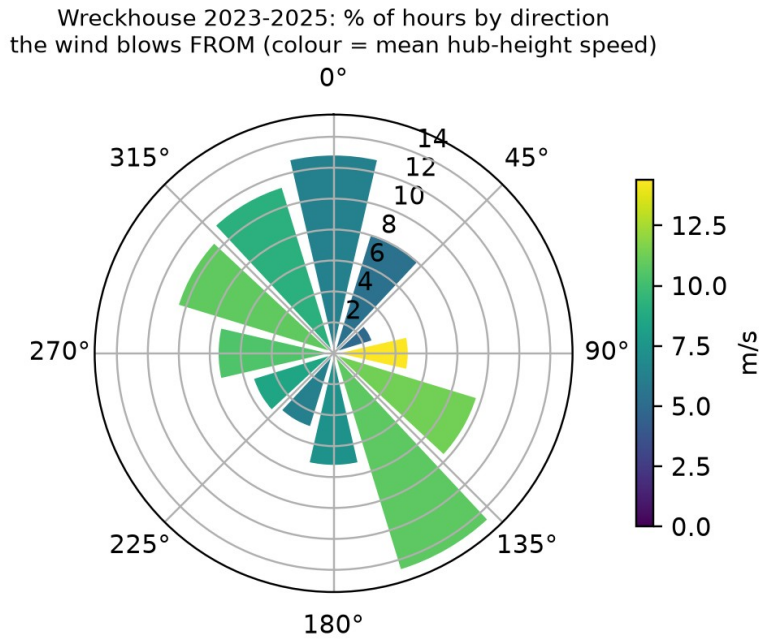
WakeWise NL

Wake-aware wind-turbine siting for Newfoundland's west coast: real wind, real terrain, QAOA in Qiskit, run up to 24 qubits on IBM Quantum System One in Quebec

Team AI Entanglers

Mostafa Wasib · Mahmoud Al-Moghrabi

Qiskit Fall Fest 2026 · Memorial University



Real input: 26,264 hours of wind at Wreckhouse, NL, 2023 to 2025

WakeWise NL in one minute

154 GWh/yr

from the recommended 9-turbine layout: the electricity of ~9,100 NL homes

~340x

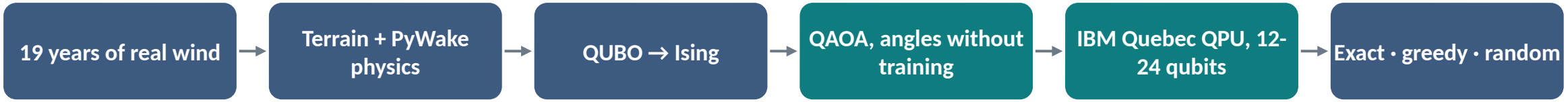
more often than random guessing, QAOA returns the optimal layout (12 qubits, noiseless simulation)

24 qubits

on IBM's Quebec machine: QAOA closes 66% of the gap to the best layout, and the exact optimum appeared (chance: 1 in 17 million)

14 of 19 years

real wind years (2007-2025) in which the recommended layout is the best possible; a one-year design: 5 of 19



Teal = quantum steps. No quantum advantage is claimed: at this size we can check every answer exactly.

Strong NL wind meets a proposed hydrogen farm

Wreckhouse, southwest Newfoundland

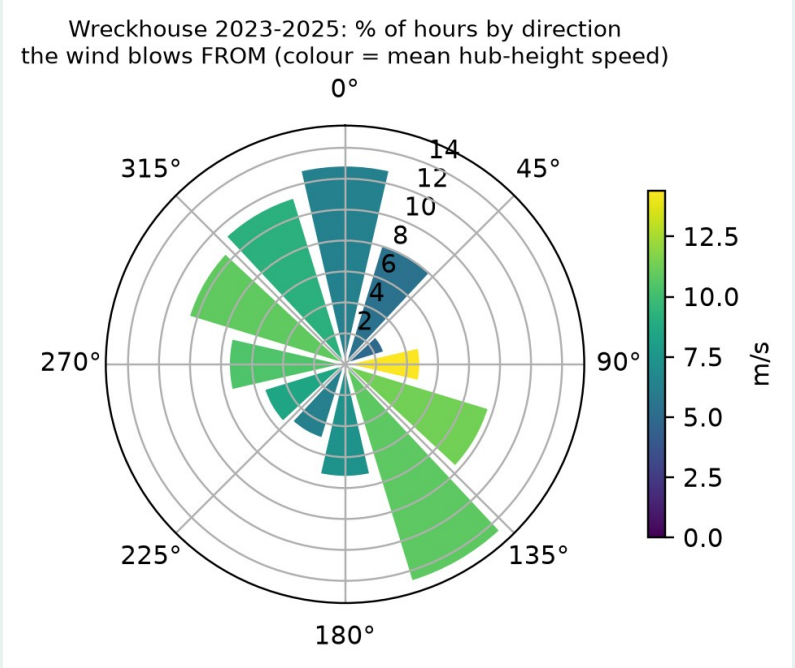
Gales here once blew trains off the tracks. Over 2023 to 2025 the wind at a 110 m hub averaged 9.0 m/s: a 4 MW turbine would run at a 50% capacity factor.

Project Nujio'qonik, next door

150+ turbines proposed on the Port au Port Peninsula and in the Codroy Valley to make hydrogen in Stephenville. Released from provincial environmental assessment, with conditions, in April 2024.

MUN's CREATE-SWIFT program asks for this

Benchmark wind-energy optimization: classical versus quantum or hybrid methods, on accuracy, cost, scalability and real-world relevance.



Bimodal wind: south-easterly gales and north-westerlies. Turbines lined up on that axis shade each other most.

Sources: ECCC hourly climate data (Wreckhouse, ID 8404343); NL Government EA bulletin, 9 Apr 2024; CBC News; CREATE-SWIFT talk, Qiskit Fall Fest 2026.

Wakes waste wind, so siting is a real decision

17,600 MWh/yr

from one 4 MW turbine standing alone at Wreckhouse (2023-2025 wind, DTU PyWake)

up to 2.4%

of that lost to one badly placed neighbour's wake; a dense 12-turbine block loses 4.2%

32% of hours

at full power, when a wake costs almost nothing: why wakes here are milder than at an average site

Decision and intended use

Which candidate sites get a turbine (bit = 1) or not (bit = 0). Output: the best layouts, the energy at stake, and whether the choice holds up year to year; for early-stage screening, not construction advice.

Sustainability benefit

More clean electricity, or green hydrogen, from the same land, steel and concrete; or the same energy from fewer turbines and less disturbed ground. The prototype shows the method; real benefit needs site-specific terrain, turbines and costs.

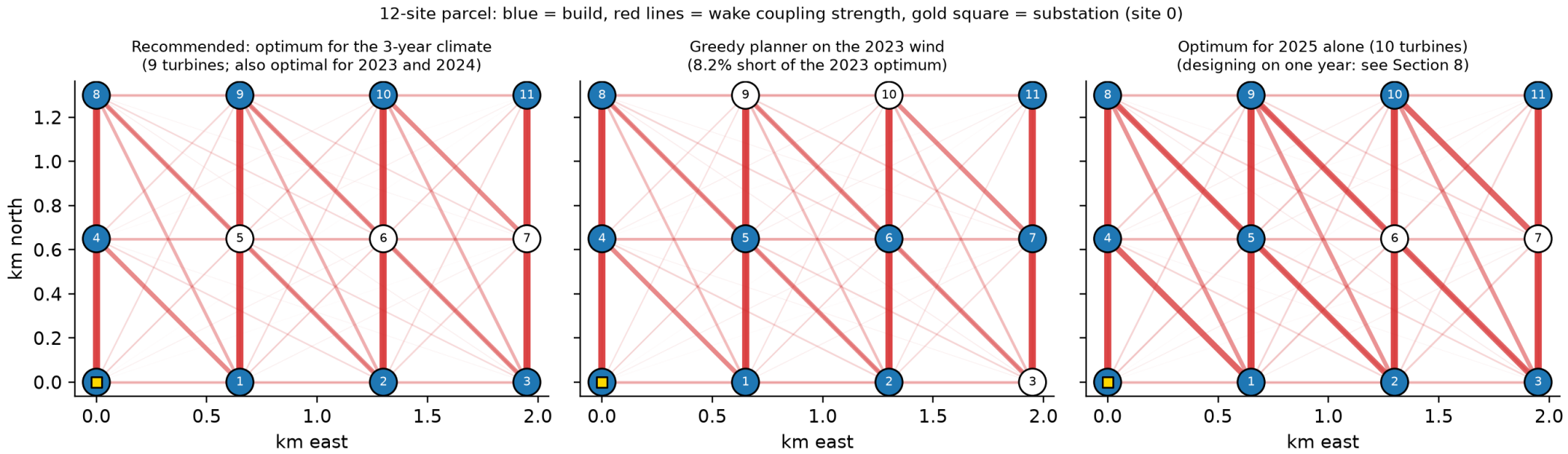
Inputs, output and scope

Input	Value used	Status
Hourly wind direction and speed	Wreckhouse, ECCC: 2023-2025 climate (26,264 h); 2007-2025 for robustness (163,068 h)	**Observed**
Hub-height wind	10 m to 110 m with power-law shear, alpha = 0.14	Assumption
Wind-farm physics	DTU PyWake: Jensen wake model; checked with full-farm and Gaussian models	**Industry-standard model**
Turbine	4 MW, 130 m rotor, power and thrust curves from PyWake's generic model	Illustrative class
Parcel and costs	Flat grid, 650 m apart; cost = 88% of a turbine's output + 2% per 650 m of cable	Illustrative, sensitivity tested
Real terrain parcel	Copernicus elevation; DTU Global Wind Atlas speed-ups on the station's real wind; sites chosen by rule	**Observed + modelled (screening)**
Output	Build / no-build bitstring = one layout	
Scope	12-16 sites on the flat parcel (19 years, wetland scenario); 12 to 24 sites on the terrain parcel; simulation, noise model and real hardware	

Wake interactions make the choice combinatorial

$$f(x) = -\Sigma\,(E - c_i)\,x_i \;+\; \Sigma\,W_{ij}\,x_i\,x_j$$

E: one turbine's energy · c_i: annualized cost as energy · W_{ij}: energy the pair loses to each other's wakes (PyWake) · MWh/yr, lower is better



Greedy (add the best site until none helps) finds the 3-year optimum, but misses the best layout in 16 of 19 single years (up to 8% short). Every layout is checked exactly against all 4,096 options.

The wake physics is already an Ising model

$$x_i = (1 - z_i) / 2$$

$$H = \Sigma\, h_i Z_i + \Sigma\, J_{ij} Z_i Z_j$$

Turbine value becomes a Z field; the wake loss between two turbines becomes a ZZ coupling.

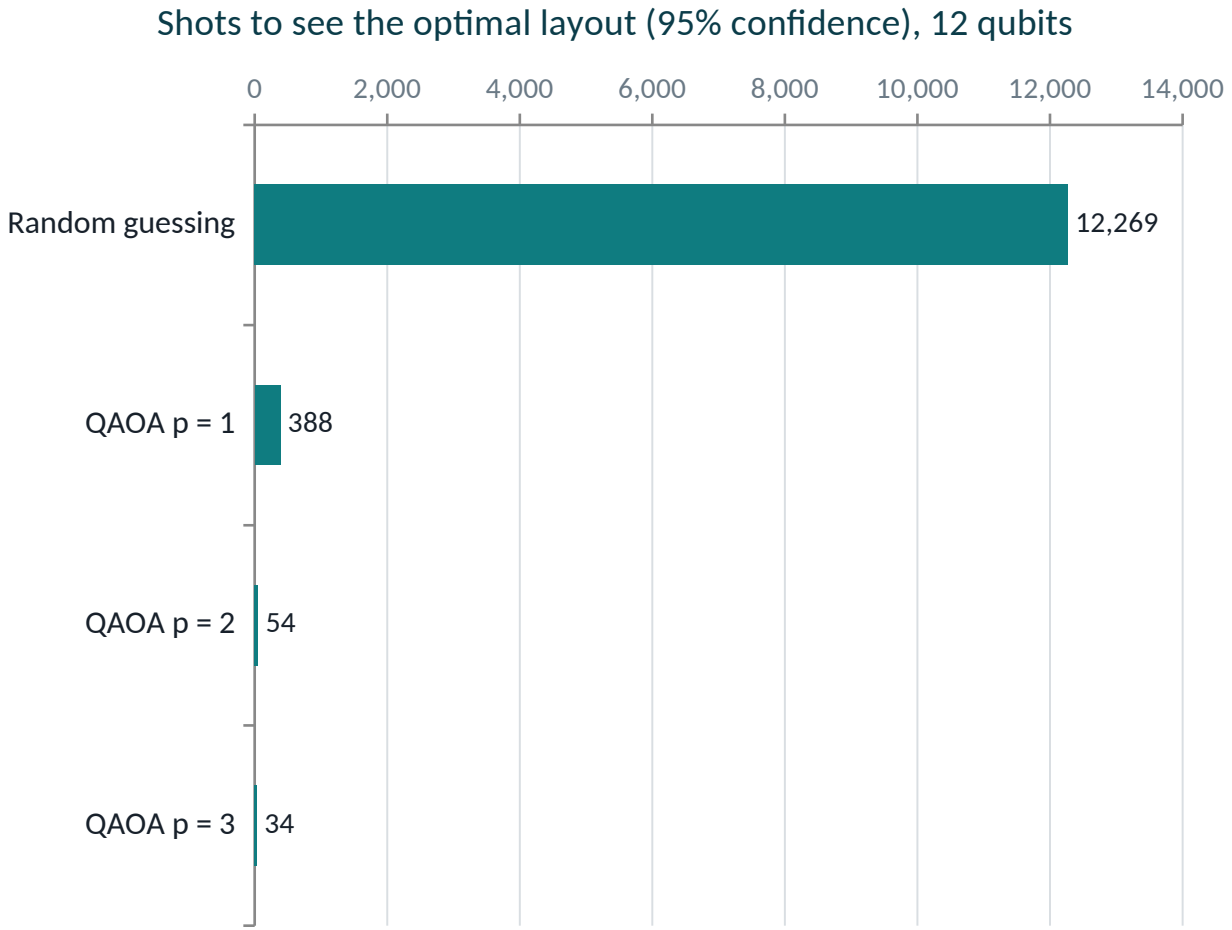
Why QAOA fits

- Natively quadratic and binary: no encoding overhead, no penalty terms
- NP-hard in general (contains maximum-weight independent set)
- CVaR training: optimize the best 25% of outcomes, because a planner needs the best layout
- Small enough to check exactly, so quality is measured, not assumed

1. **Map** *classical*
   ECCC wind → PyWake → QUBO → SparsePauliOp → qaoa_ansatz, depth p = 1 to 3
2. **Optimize** *classical*
   Transpile for IBM Heron (level 3); drop weak couplings to save two-qubit gates
3. **Execute** *quantum*
   Angles trained in exact simulation (CVaR), or set without training at 20-24 qubits; sample on ibm_quebec
4. **Post-process** *classical*
   Score every sampled layout with the full QUBO; compare with exact, greedy and random

QAOA finds the best layout ~340x more than chance

Noiseless simulation



8.4% vs 0.02%

of shots return the optimal layout (QAOA p = 3 vs random guessing)

34 vs 12,269

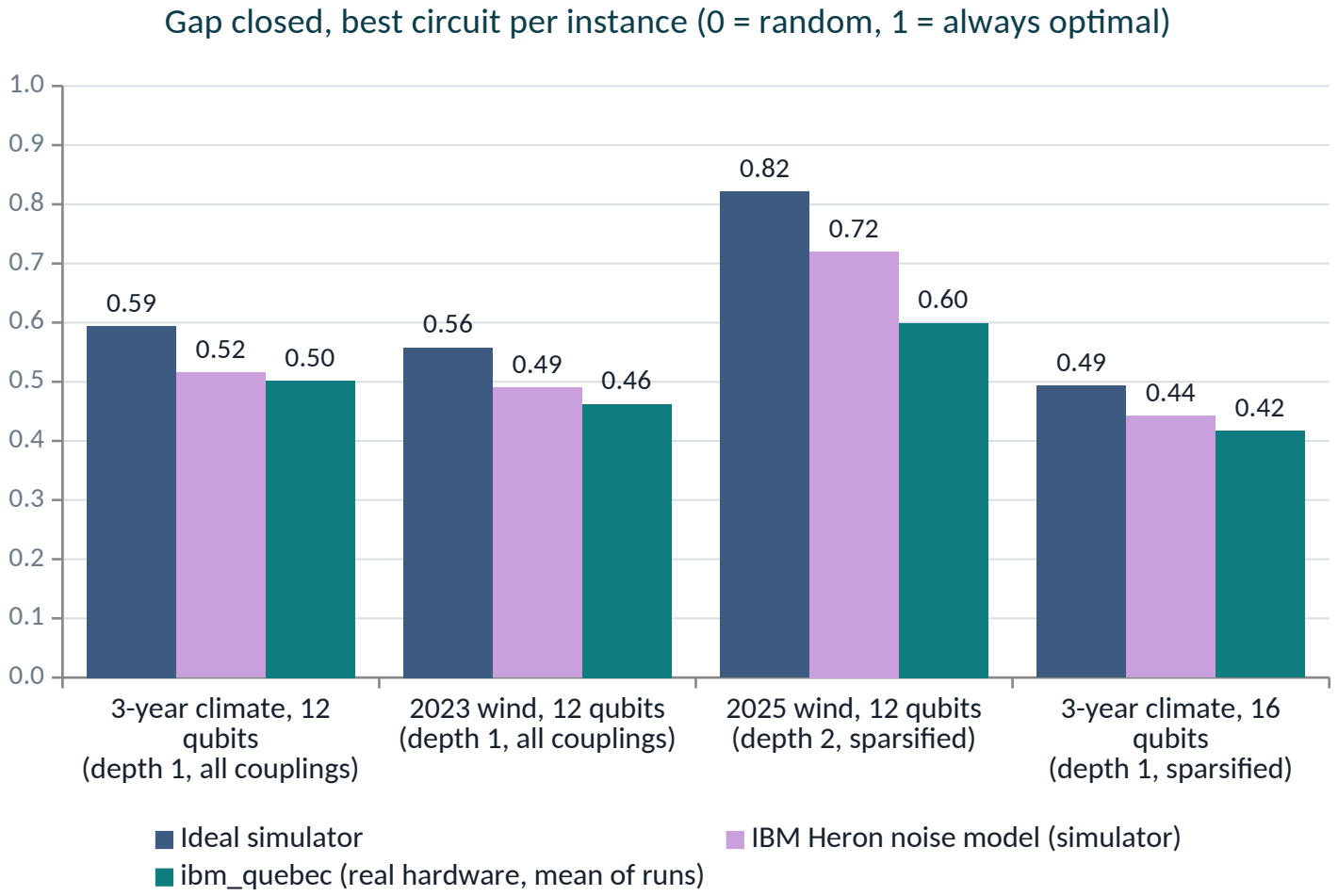
shots needed to see the optimum with 95% confidence

20%

of QAOA samples (p = 2) beat greedy, on average, in the 16 single years where greedy misses (19 years tested)

Honest caveat: a typical QAOA sample (gap closed 0.76) is below the best layout; QAOA's value is in its best few dozen shots.

On IBM's Quebec machine, the signal survives



Simulator + real hardware

ibm_quebec (IBM Quantum System One, PINQ², Bromont, Quebec), 7 Nov 2026; median CZ error 0.15%; 21 jobs, 174 s of QPU in total.

3-year climate, 4 runs: the optimum in 0.47% to 0.63% of shots, 23x random.

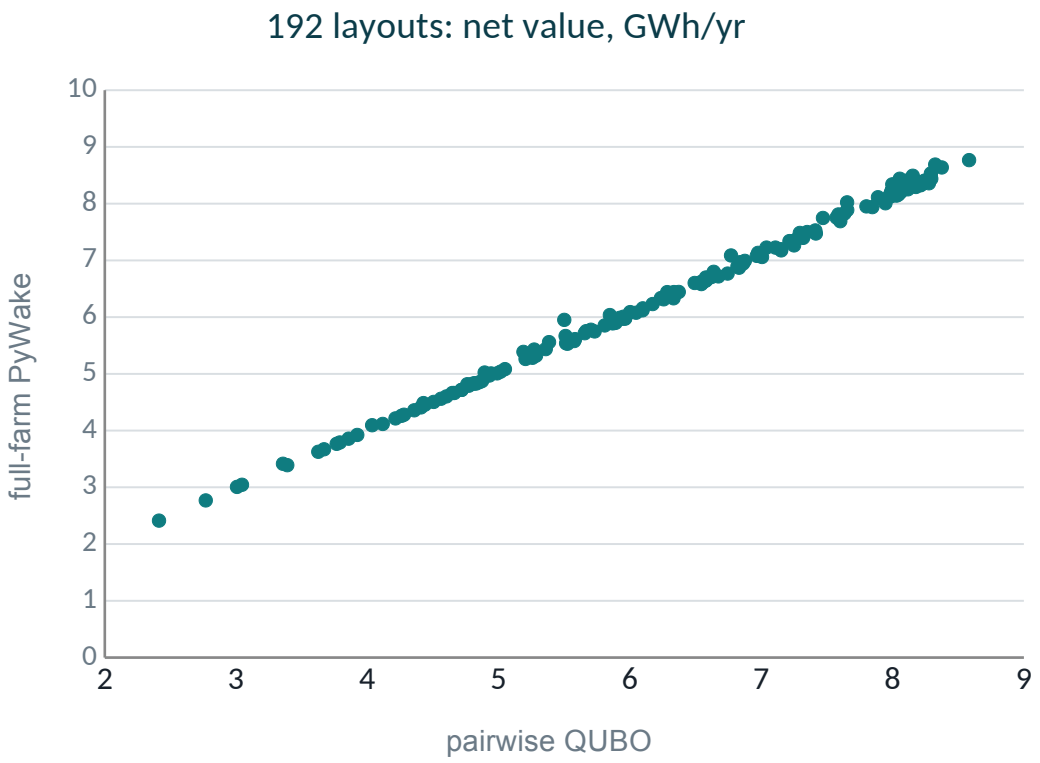
2023 wind, 4 runs: 15% of hardware samples beat the greedy planner (random guessing: 2%).

2025 wind, 1 run: the optimum 68x more often than random.

We found a faulty qubit: every depth-2 run sat on qubit 101, which misread 1 as 0 in 65% of shots (hence readout mitigation's big gain there). On healthy qubits depth 2 did no better (gap 0.38): its extra gates are the real limit.

Control circuits (should be random) score -0.02: the gain is QAOA, not noise.

Validated: full simulations and 19 years of wind



Rank correlation 0.993 with full-farm PyWake and 0.992 with a Gaussian wake model; the QUBO's best layout ranks #1 and #1.

Layout designed on	Best in	Avg. loss	Worst year
2023-2025 (recommended)	**14 of 19**	**19 MWh/yr**	**167 MWh/yr**
2025 wind only	5 of 19	104 MWh/yr	416 MWh/yr

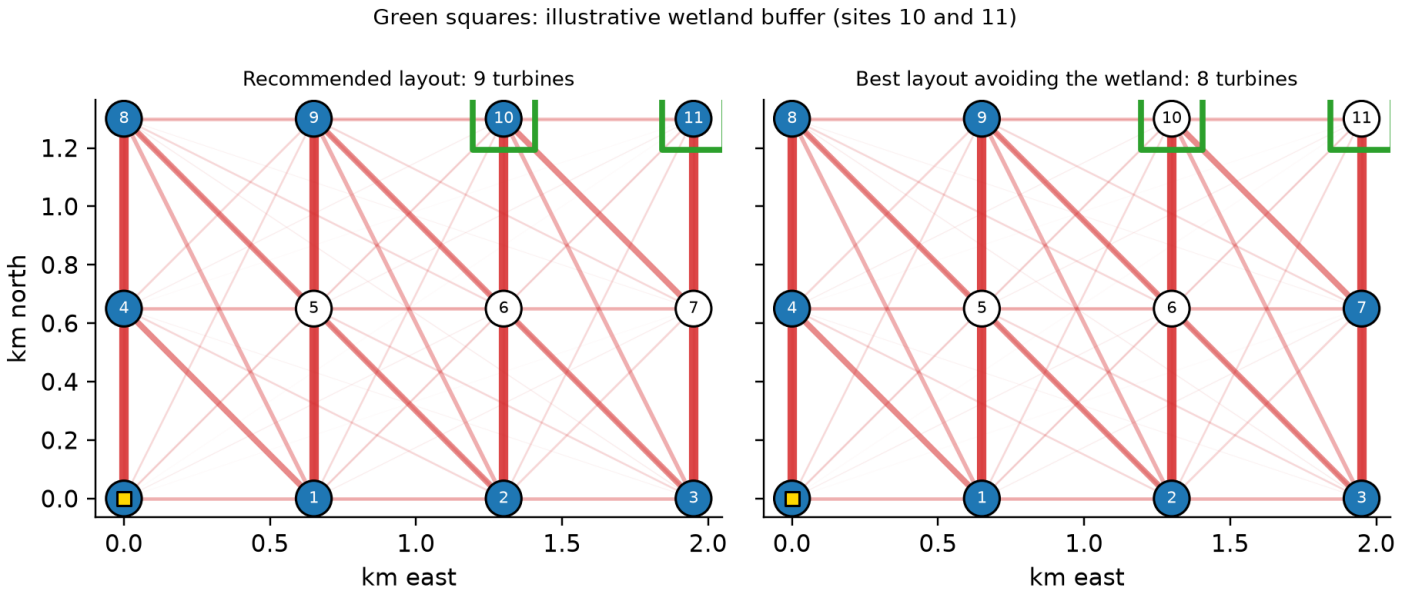
16 of 19 years

in which greedy misses that year's best layout (up to 8% short)

20%

of QAOA samples (p = 2, noiseless) beat greedy in those years, on average

Protecting a wetland: what it costs



Illustrative scenario: two sites fall inside a wetland buffer. Next to Wreckhouse, the Grand Codroy Estuary is Newfoundland's only Ramsar wetland of international importance.

519 MWh/yr

of net energy is the price of keeping turbines out of the wetland

9 → 8 turbines

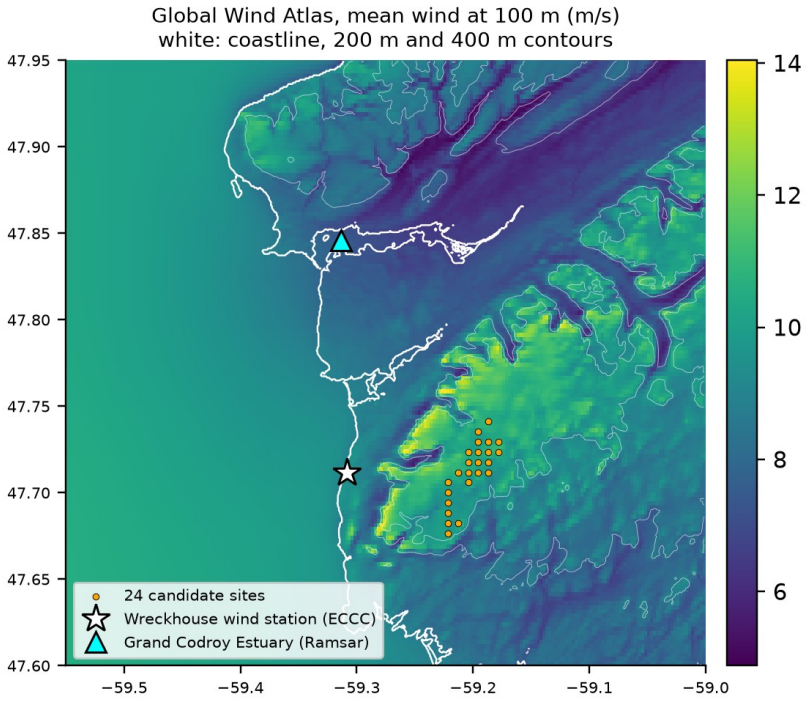
the optimizer re-plans the layout instead of just dropping two sites

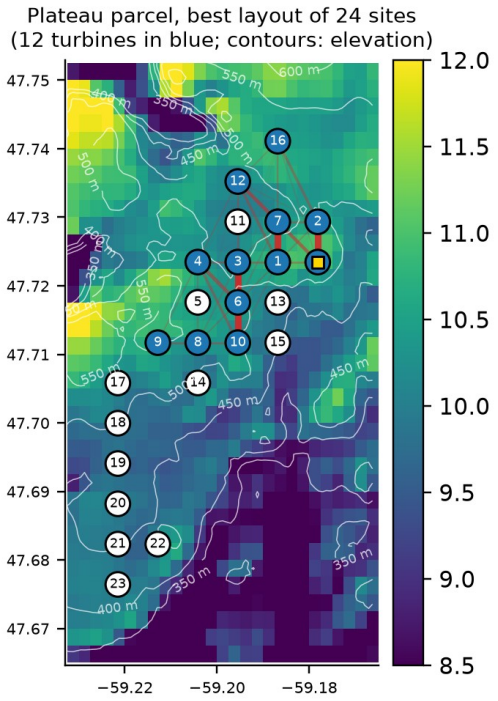
3.2% of shots

QAOA (p = 3, noiseless) finds the new optimum, 133x random; the wetland is just a stronger Z field

A real parcel on the plateau above Wreckhouse

Screening level





Sites chosen by rule: on land, slope ≤ 10%, terrain as rugged as the station's (WAsP's RIX test), 3 km from the Ramsar estuary; grown from the windiest point. Wind: the station's real hours scaled by DTU's Global Wind Atlas.

423-550 m

24 plateau sites, 9.7-10.9 m/s in the atlas (station 9.5; our series 9.0, -5%)

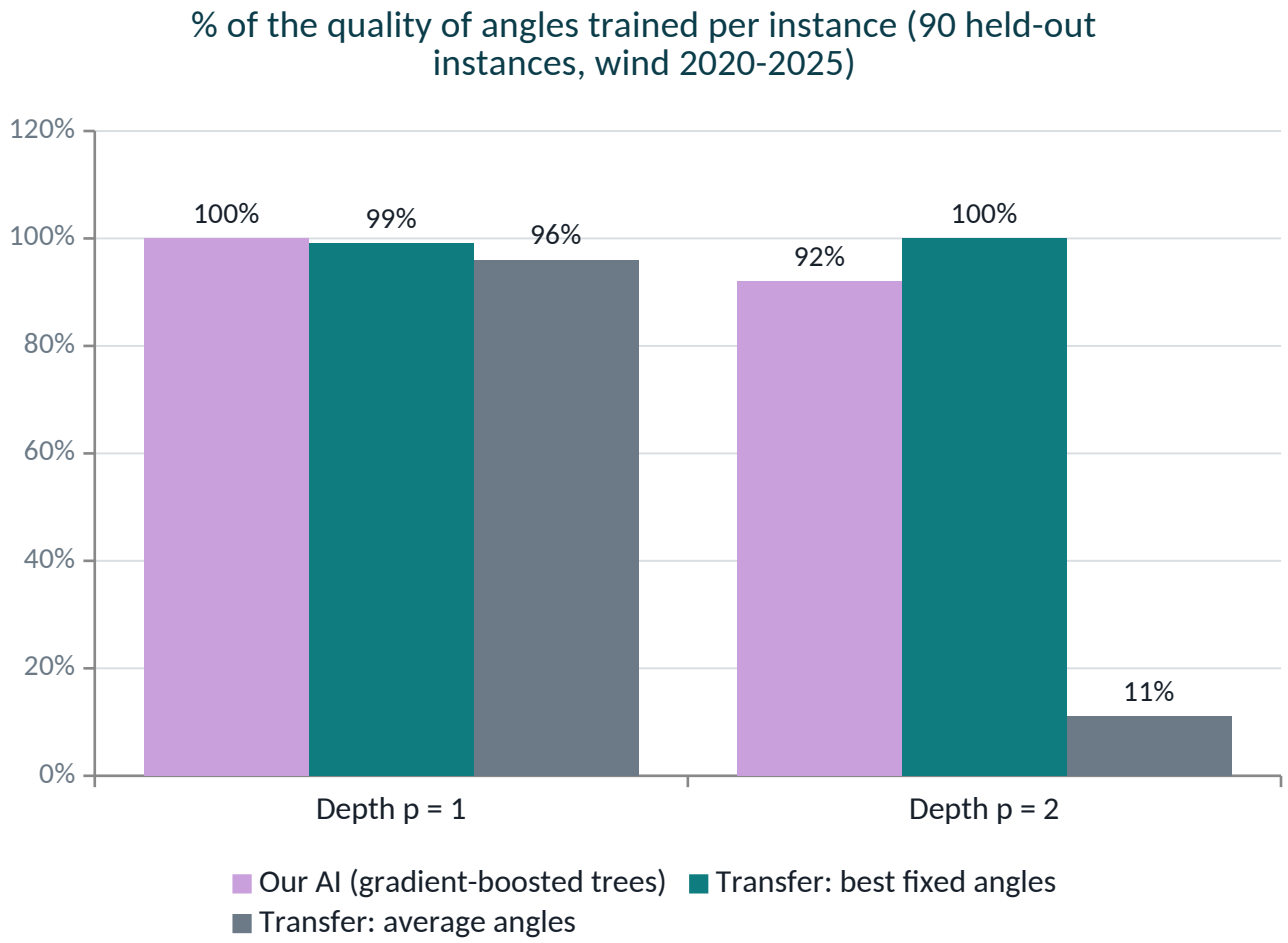
12 vs 4 years

best possible when designed on all 19 years vs on 2023-2025: on terrain, design on the long record

804x

QAOA (p = 2, noiseless) finds the 16-site optimum more often than random

AI sets the angles, but transfer works too



Noiseless, held-out years

100% / 92%

our AI at depth 1 / depth 2: no training loop needed

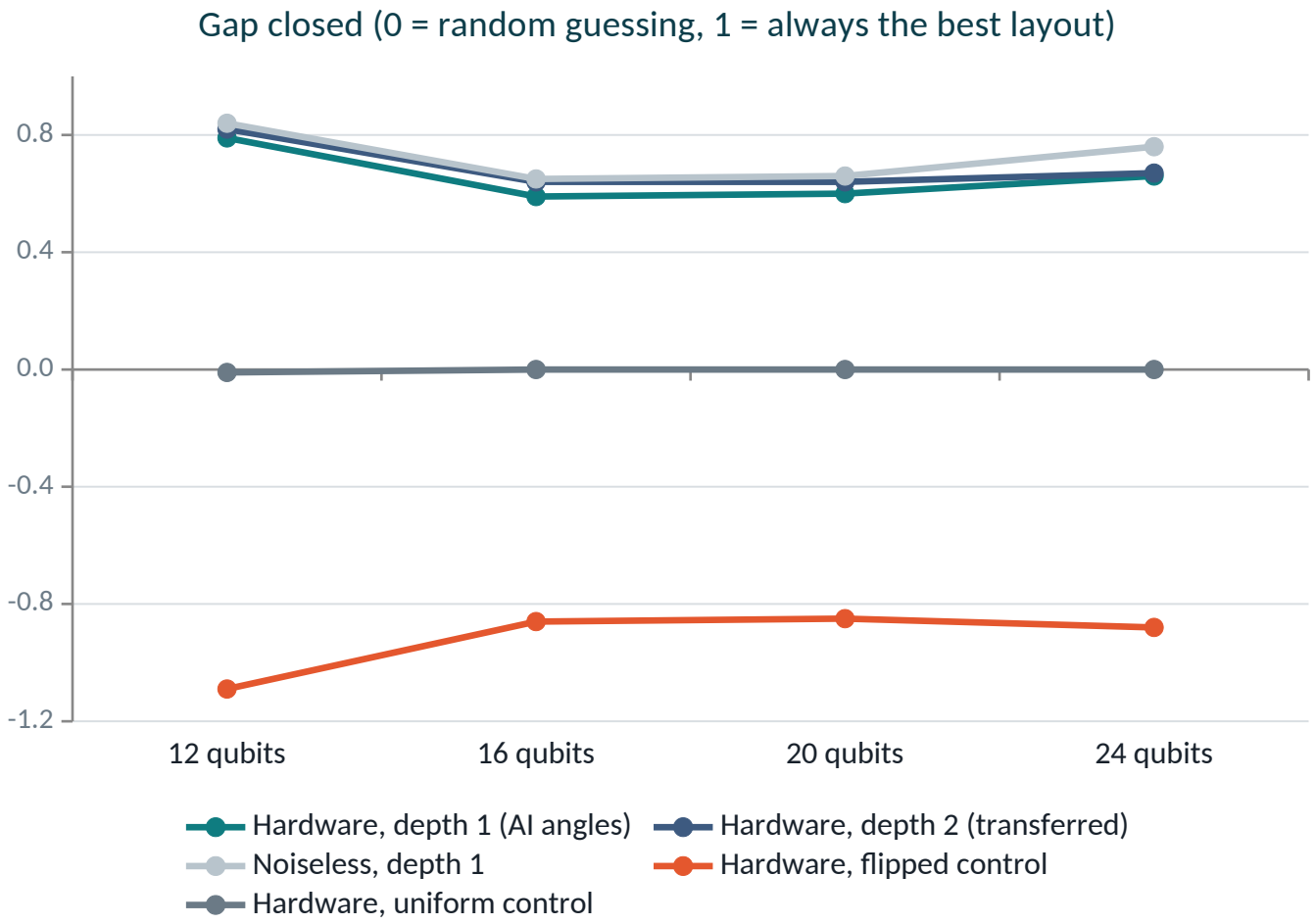
100%

one transferred angle set at depth 2: it beats our AI, an honest negative result

11%

averaging angles at depth 2 fails: they sit in different basins

24 qubits on IBM's Quebec machine



Real hardware

0.66 vs 0.76

gap closed at 24 qubits on hardware vs noiseless (213 two-qubit gates)

4 in 24,000

shots returned the exact best of 17 million layouts; chance: 1 in 17 million

-0.88

the flipped control: same gates, one sign changed, so it steers away; the device follows the physics

What the evidence supports, and what it does not

Supports

- Siting is combinatorial on real NL wind; the pairwise QUBO matches full wind-farm simulations
- The recommended layout is the best possible in 14 of 19 real wind years
- QAOA concentrates on the best layouts, from ~340x random (12 qubits, noiseless) to a clear signal on hardware at 24 qubits
- Angles transfer: no training loop is needed at 20-24 qubits (our AI works, transfer works as well)
- Matched controls behave as physics predicts: flipped circuits score below random

Does not support

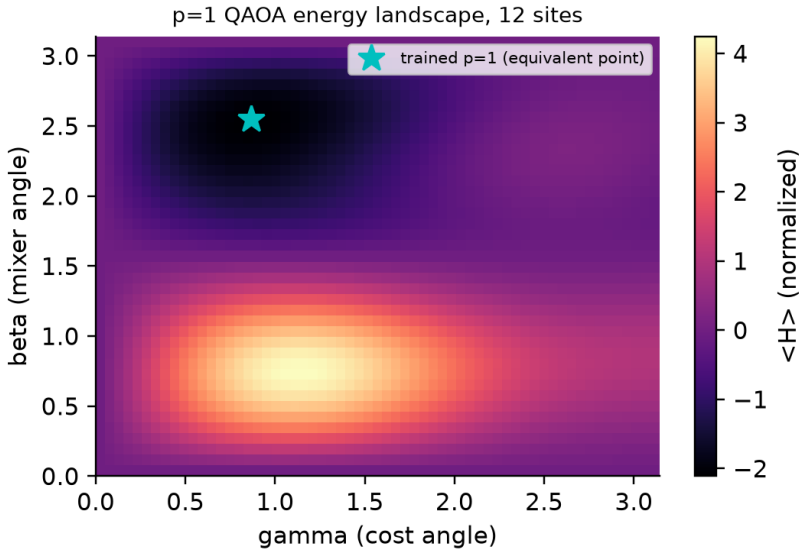
- Quantum advantage: a laptop still solves 24 sites exactly
- Bankable siting: the terrain parcel is screening-level (Global Wind Atlas); costs are illustrative
- That our AI beats simple parameter transfer: it does not
- That depth 2 beats depth 1 on today's hardware: re-run on healthy qubits, it did not

Next: measure on the plateau and model the flow site-specifically · spacing and habitat constraints as QUBO terms · constraint-preserving mixers · larger parcels against simulated annealing and MILP · partner through MUN CREATE-SWIFT and Mitacs Accelerate

Resources: notebook about 30 minutes on a laptop; 21 hardware jobs, 174 s of QPU in total.

More clean energy from the same wind

- Real Wreckhouse wind, real terrain and DTU PyWake physics give a QUBO that matches full wind-farm simulations
- The recommended layout is the best possible in 14 of 19 real wind years, and protecting a wetland costs a known 519 MWh/yr
- QAOA in Qiskit finds it ~340x more often than chance, and on IBM's Quebec machine it keeps a clear signal up to 24 qubits, with angles set without training



p = 1 QAOA energy landscape for the 12-site parcel; the star marks the trained angles.

github.com/mrw-soumik/Qiskit-AI-Entanglers-

Thank you. Questions?

Credits: QAOA workflow adapted from Tinh Bui's workshop notebook; IBM Qiskit 101 and the Fall Fest hardware example; DTU PyWake; ECCC data (Open Government Licence - Canada); DTU Global Wind Atlas (CC BY 4.0); Copernicus DEM (ESA). Full credits in the notebook.